DISTRIBUTED SMART TRAFFIC LIGHT CONTROL SYSTEM

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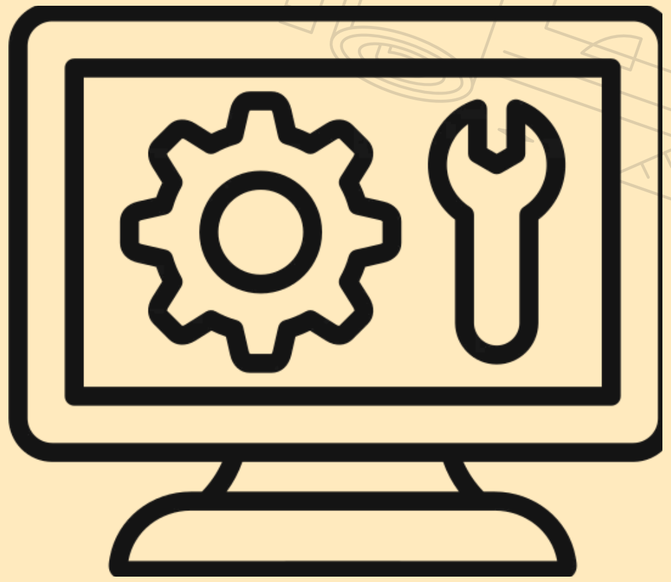
INTRODUCTION

Project goal: To create a decentralized traffic management system where each intersection autonomously adapts to traffic flow and cooperates with neighboring ones to eliminate congestion and delays in real time.

CHOICE OF TOOLS

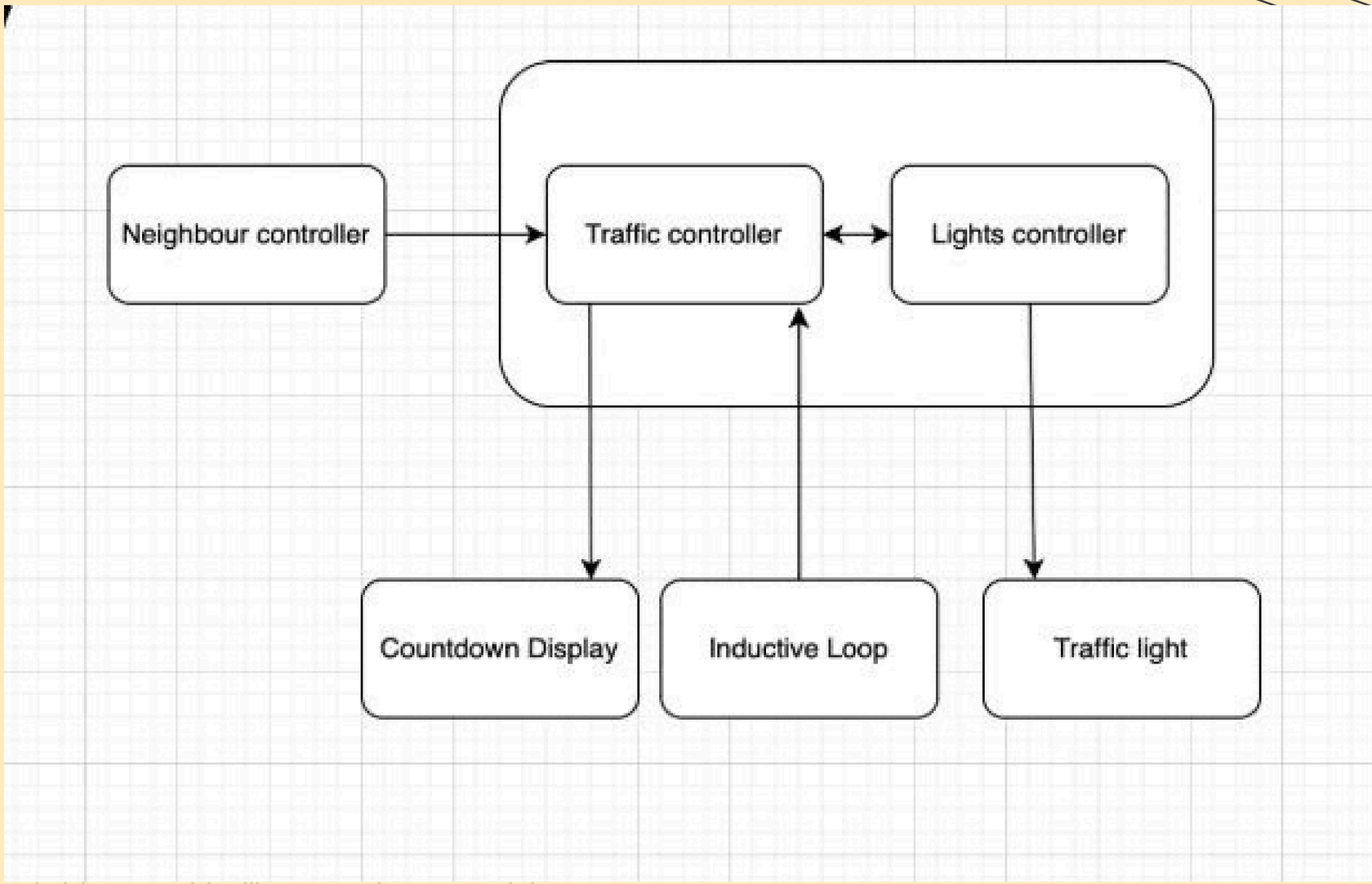
1. VSCode
2. Node.js
3. MQTT
4. JSON
5. Grafana
6. Influxdb
7. Docker



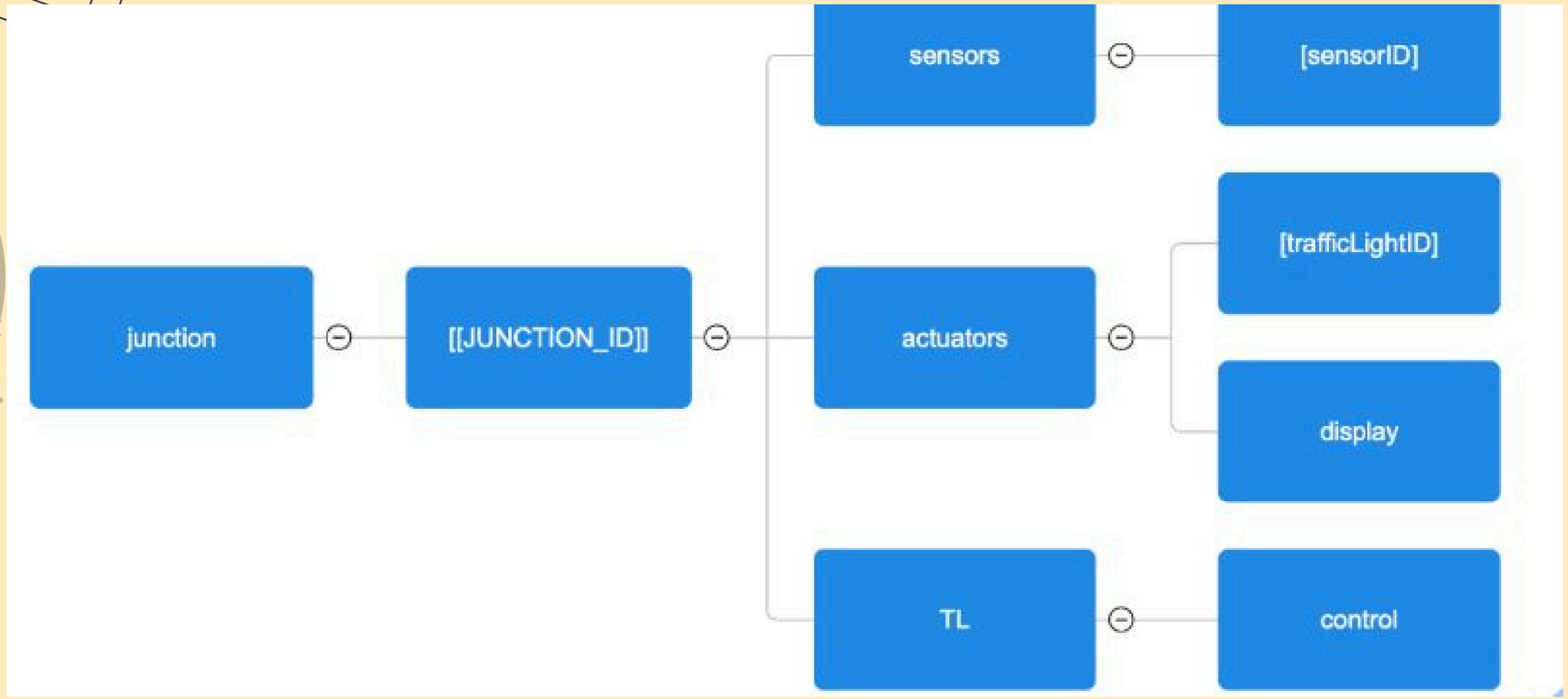
IMPLEMENTED CONTROLLERS

COMPONENTS COMMUNICATION

(all via mqtt)

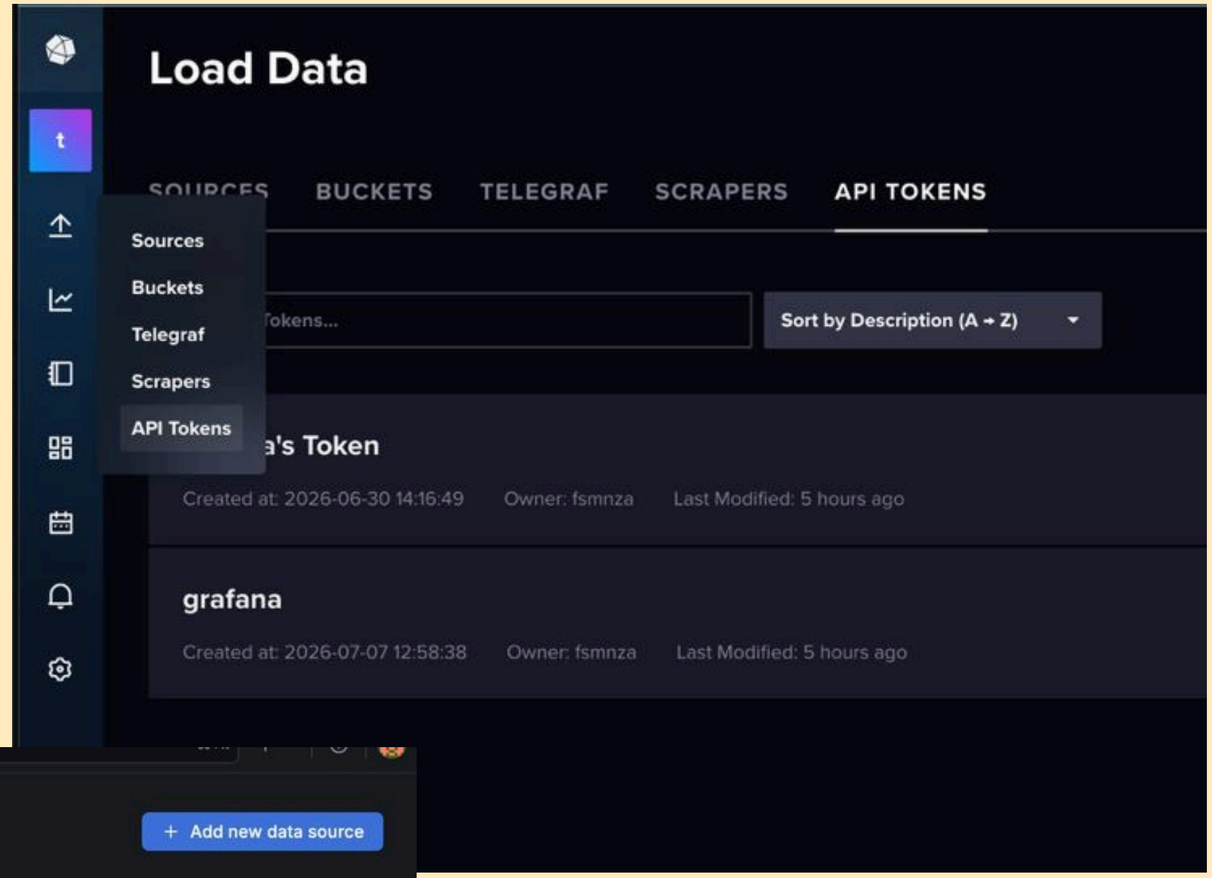


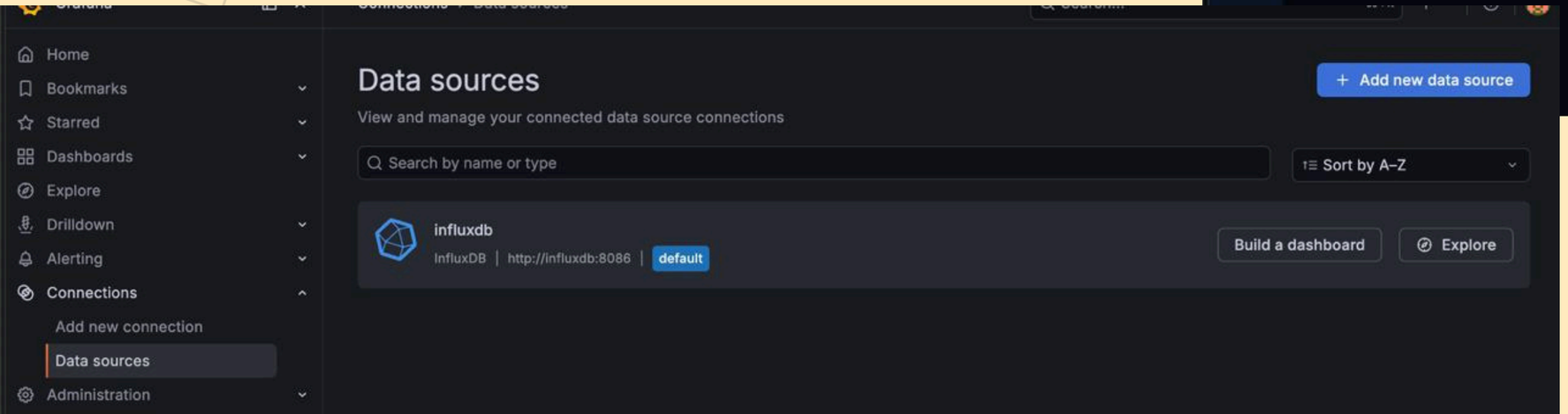
MQTT TOPICS


junction
[[JUNCTION_ID]]
sensors
actuators
TL
[sensorID]
[trafficLightID]
display
control

INFLUXDB & GRAFANA

Get InfluxDb Token

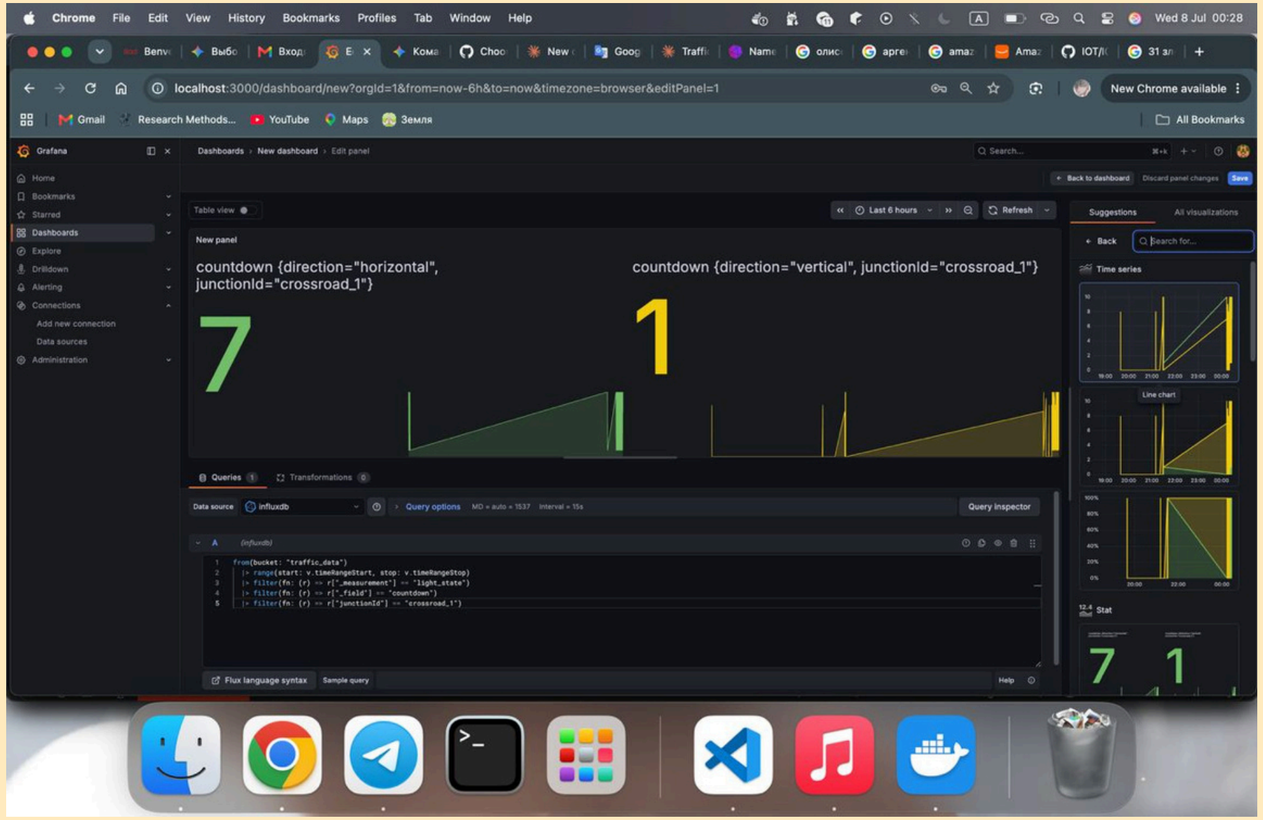




Add data source

INFLUXDB

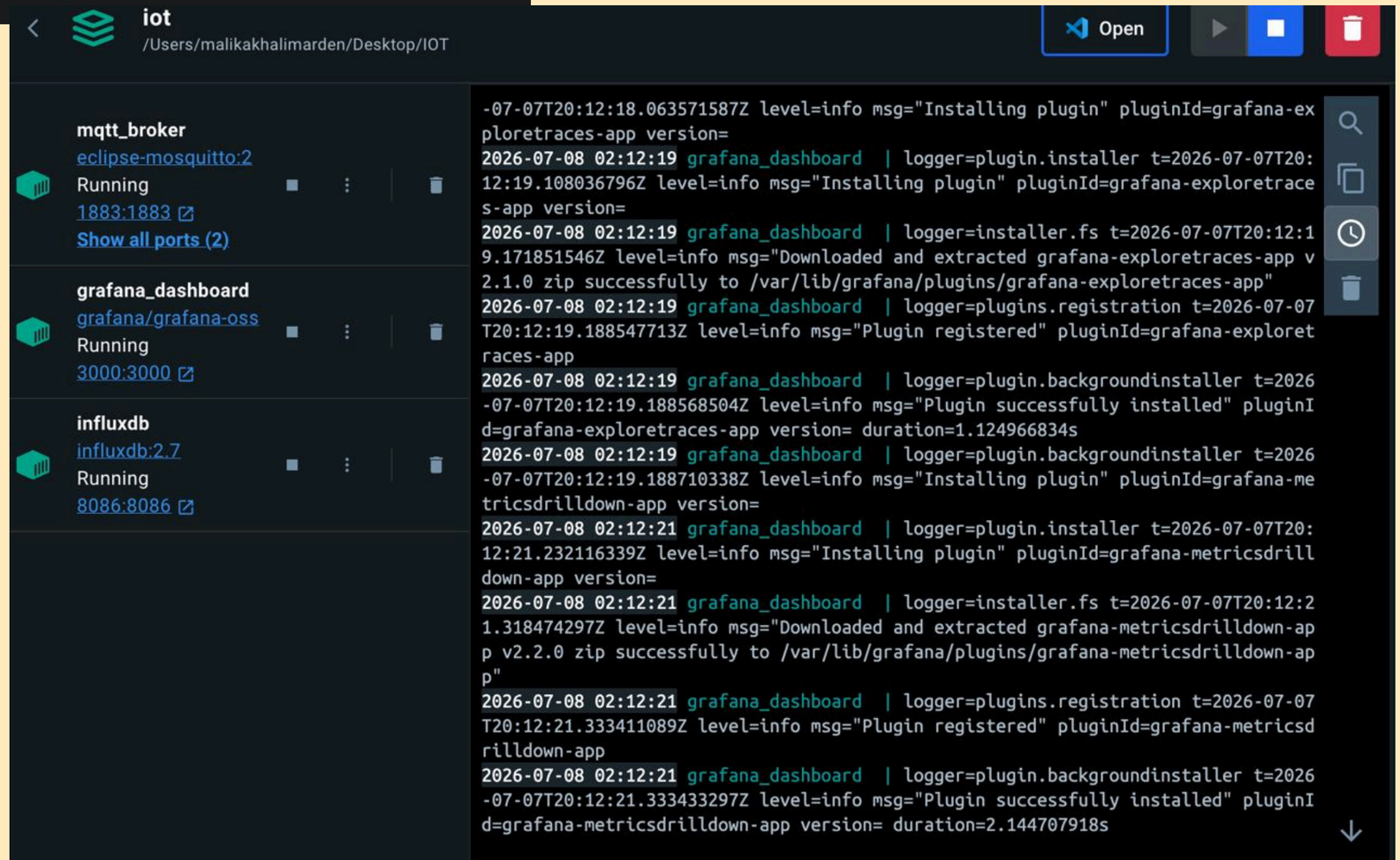
GRAFANA



DOCKER

Upped the docker

```
malikakhalimarden@Malikas-MacBook-Air IOT % docker compose up -d
[+] Running 4/4
 ✔ Network iot_default          Created
 ✔ Container mqtt_broker        Started
 ✔ Container influxdb           Started
 ✔ Container grafana_dashboard  Started
```



SIMULATION

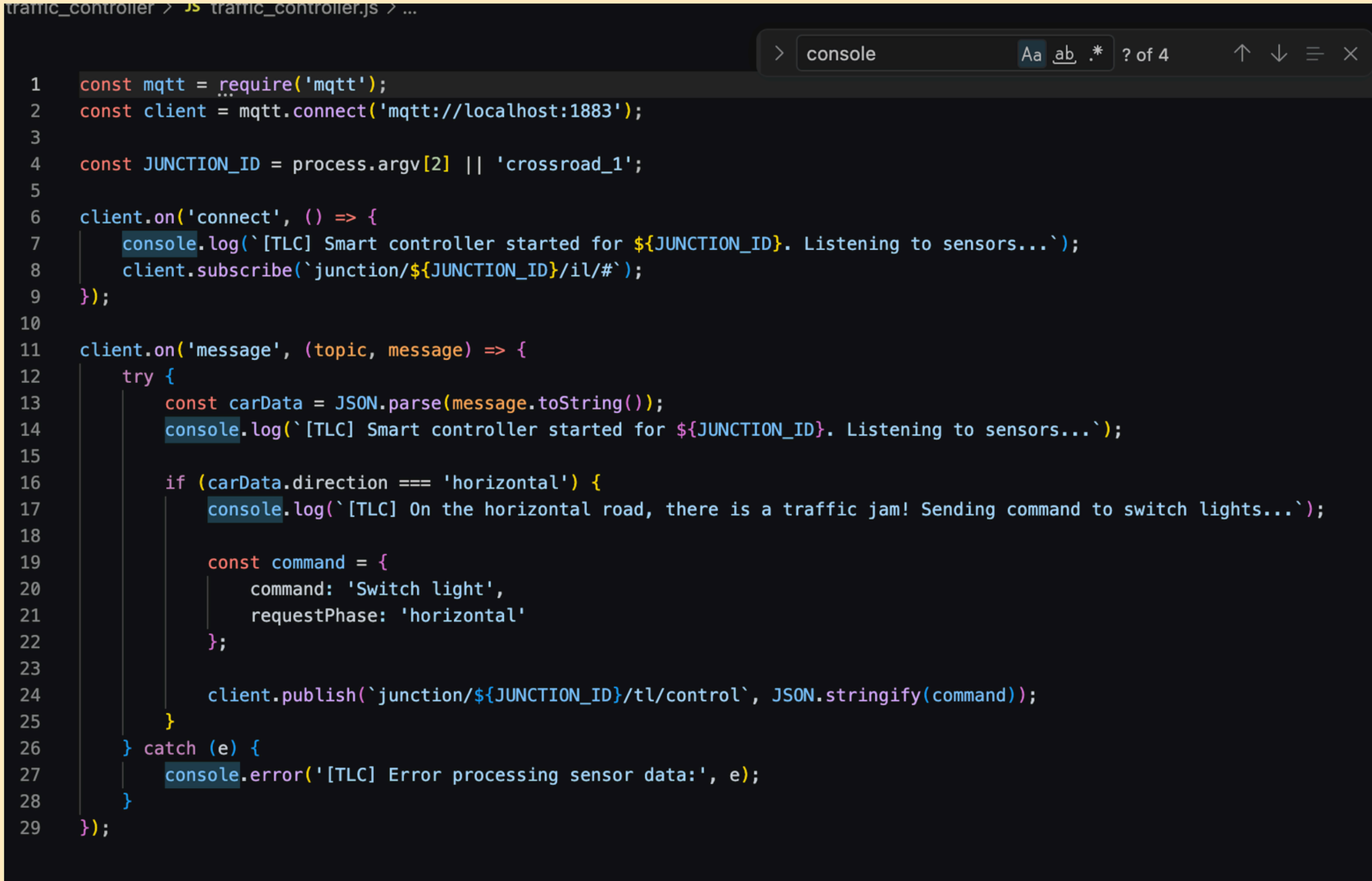
```
1
2   const { spawn } = require('child_process');
3   const path = require('path');
4
5   const junctionIdArg = process.argv[2] || 'crossroad_1';
6
7   const scripts = [
8       { name: 'TLC', path: 'traffic_controller/traffic_controller.js' },
9       { name: 'Crossroad', path: 'crossroad_1.js' },
10      { name: 'DBLogger', path: 'db_logger/db_logger.js' },
11      { name: 'IL', path: 'inductive_loop/cars_over_il.js' },
12  ];
13
14  const runningProcesses = [];
15
```

console

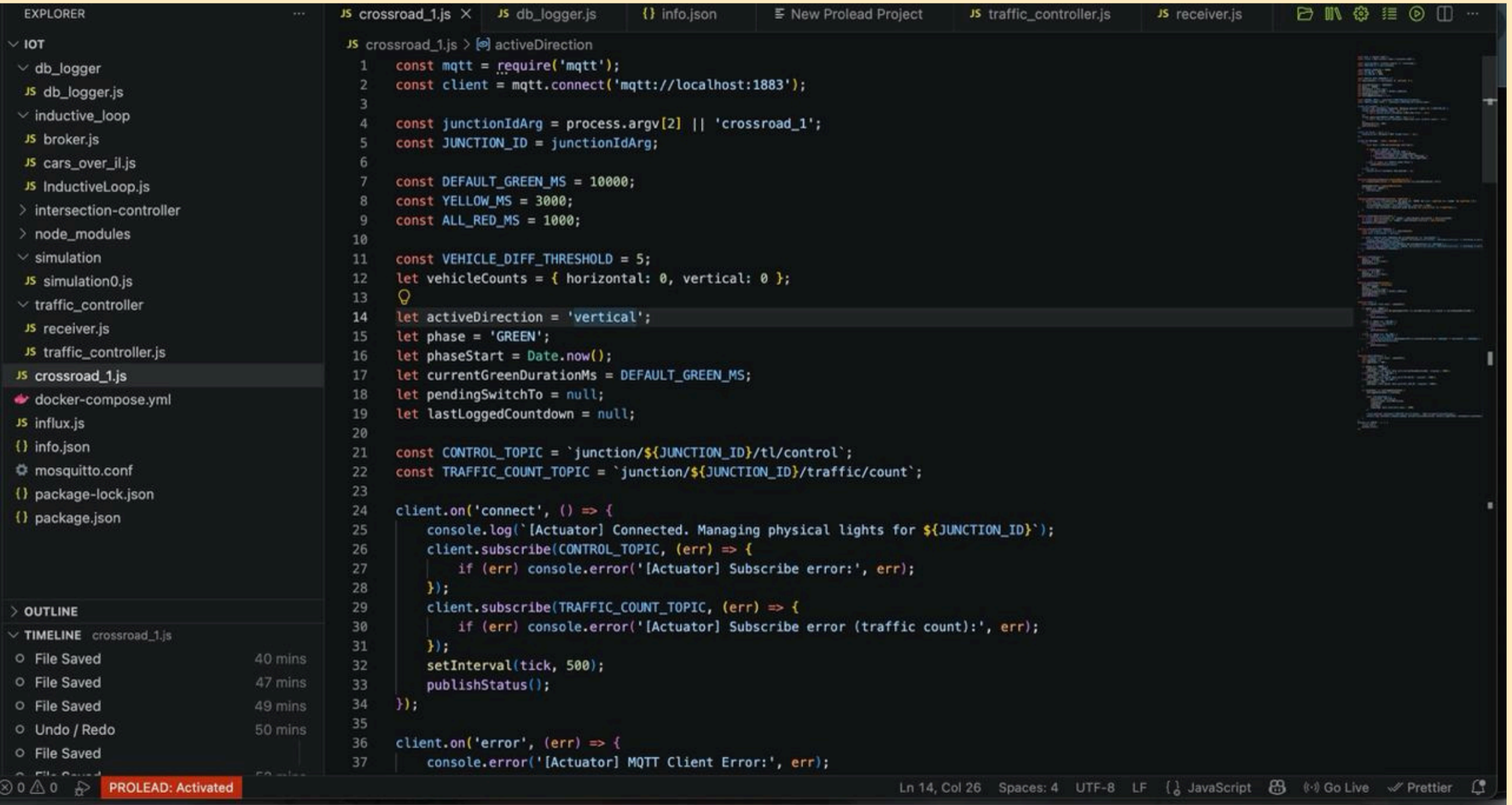
We have simulation0.js. And we just run it. And it runs another js class

TRAFFIC

Traffic is the smart traffic light switches phases using sensors.



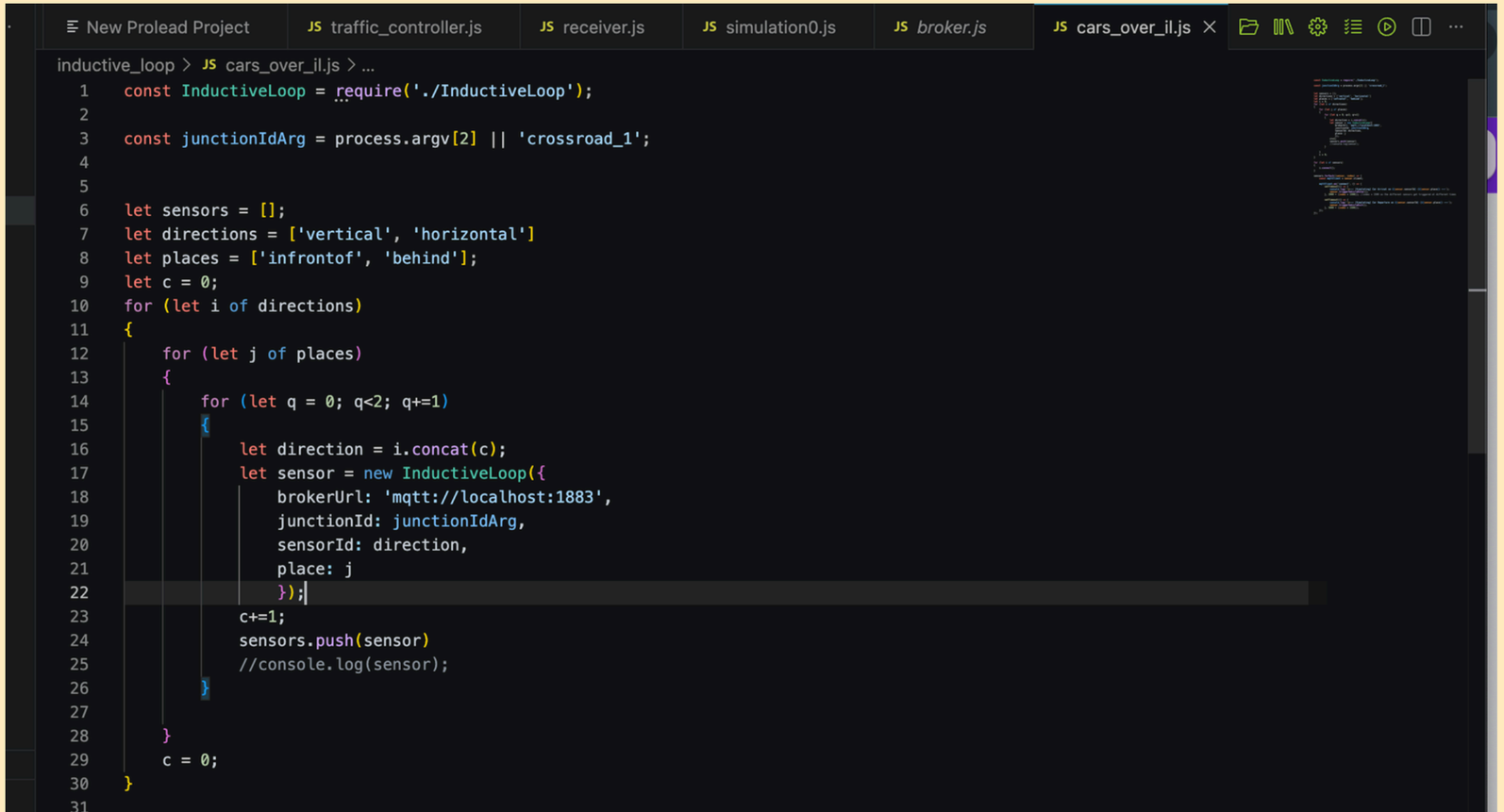
CROSSROAD



The traffic light machine switches phases according to traffic jams.

CARS_OVER_IL

Simulation of vehicle motion sensors at an intersection.



THANK YOU